



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical
Sciences Chennai- 602105



Student Name: N. Prince Raj

Reg. No.:192312284

Course Code: ECA1420

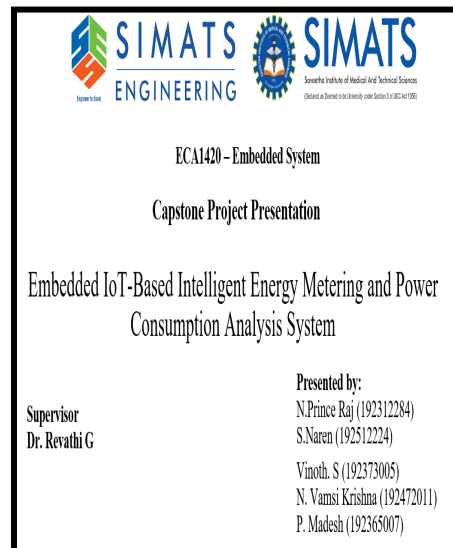
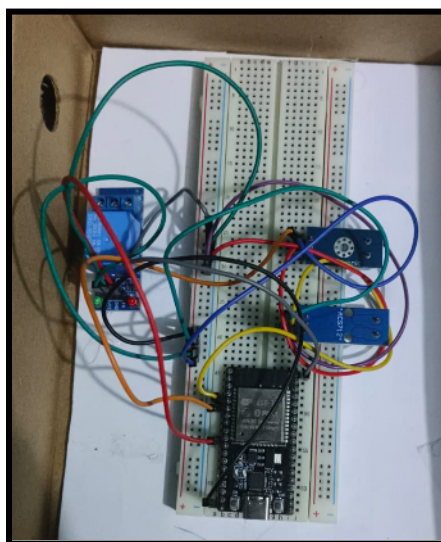
Slot: B

Course Name: Embedded System

Course Faculty: Dr.G. Revathi

Project Title: Embedded IoT-Based Intelligent Energy Metering and Power Consumption Analysis System

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)



Project Description: The project is an Embedded IoT-Based Intelligent Energy Metering and Power Consumption Analysis System that monitors voltage, current, and power consumption using suitable sensors. An ESP32 microcontroller processes the sensor data in real time and sends it to ThingSpeak and Blynk for remote monitoring. A predefined power threshold helps identify excessive energy consumption. The system provides real-time energy analysis and helps users understand their power usage. It is a cost-effective solution for improving energy efficiency and supporting better power management.

My contribution focused on hardware assembly, sensor integration, wiring, and system testing. I assembled the required hardware components and connected the sensors, ESP32 microcontroller, and other modules according to the circuit design. I tested the individual components to ensure proper operation and verified sensor readings under different conditions. I also identified and corrected wiring and hardware-related issues during testing. This contribution helped improve the reliability, accuracy, and overall performance of the system.

Student Signature

Guide Signature